

CONNECTED ARENA

Real-Time Multiplayer Fan Engagement Platform

AWS Sports AI Innovation Cup 2026 | Challenge 4: Connected Arena

TEAM

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100%

Serverless

9

AWS Services

<20ms

WebSocket Latency

3

AI Personas

d1706ex99mjina.cloudfront.net

1/5

THE PROBLEM & OUR SOLUTION

"Watching football today is passive. Fans react — they don't participate."

No Interactivity

Broadcast/streaming offers zero participation.

Disconnected Fans

Social reactions come after the moment, not during.

Generic Experience

Every fan sees the same commentary.

↓ OUR SOLUTION ↓

PREDICT

Live Predictions

YES/NO on shots, penalties, free kicks with countdown timer + instant XP.

AI

AI Commentary

Bedrock Claude Haiku generates unique commentary per fan persona.

RANK

Real-Time Leaderboard

Global ranking updates after every goal. 5 tiers: ROOKIE → APEX.

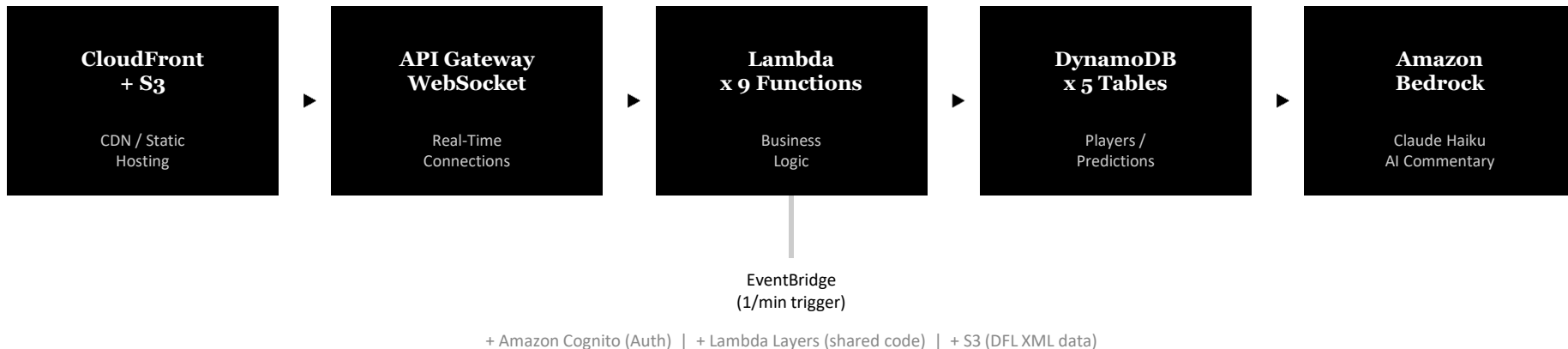
SPATIAL

Stadium Big Screen

Full-screen mode for jumbotrons and watch parties with live QR join.

AWS ARCHITECTURE — 100% SERVERLESS

9 AWS Services | 9 Lambda Functions | 5 DynamoDB Tables | Zero EC2



1. Fan visits CloudFront → opens WebSocket → API Gateway routes to connect Lambda
2. EventBridge fires event_emitter Lambda every 60s → reads next DFL match event from DynamoDB
3. Resolves open predictions (awards XP) → invokes broadcast Lambda → pushes MATCH_EVENT to all fans
4. For GOAL/PENALTY/VAR/CARD → invokes bedrock_commentary Lambda → unique AI_COMMENTARY per fan persona
5. Frontend: score updates, feed card, win-prob bar, prediction modal, leaderboard auto-refresh

SPATIAL VISION & NORTH STAR ECOSYSTEM

From a single screen to a connected fan ecosystem — matchday, mid-week, everywhere.

 MOBILE LIVE	 STADIUM SCREEN BUILT	 WATCH PARTY BUILT	 AR IN-STADIUM CONCEPT	 WEARABLE CONCEPT
• Full arena experience • Predictions + XP • AI commentary • Leaderboard + chat • Push notifications	• /stadium big-screen mode • Giant scoreboard + feed • Live crowd vote bars • QR code to join • AI commentary overlay	• Shared screen companion • Fan count + crowd votes • Projected leaderboard • Group challenges • Second-screen sync	• Phone camera AR overlay • Live stats on pitch • Player XP badges floating • Section-based challenges • Proximity fan matching	• Haptic buzz on prediction • Tap YES/NO on wrist • XP + streak notifications • Heart rate challenges • Cross-device identity

NORTH STAR: A persistent fan identity that travels across phone, stadium, wearable, and living room — every interaction earns XP, every touchpoint is connected.

LIVE DEMO & WHY WE WIN

LIVE

<https://d1706ex99mjina.cloudfront.net>

Full fan experience — join, predict, earn XP, AI commentary

STADIUM

<https://d1706ex99mjina.cloudfront.net/stadium>

Big Screen mode — spatial proof-of-concept for jumbotrons

GITHUB

<https://github.com/pacyuzul6/connected-arena>

Full source: Next.js frontend, 9 Lambdas, SAM template

WHY CONNECTED ARENA WINS

01 Personalized AI at Scale

Bedrock generates a unique message per fan per event — not one broadcast, N individual conversations.

02 True Real-Time WebSocket

API Gateway WebSocket pushes to browsers instantly. EventBridge + Lambda + DynamoDB with zero polling.

03 Complete Fan Journey

Pre-match predictions → live XP → HT AI summary → post-match standings → mid-week retention.

04 Spatial Proof-of-Concept

Stadium big-screen mode, watch-party view, QR join — plus wearable & AR concepts articulated.